

Mount Kenya University

Unlocking Infinite Possibilities

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ADM NUMBER: BSCCS/2024/56268

COURSE: COMPUTER SCIENCE

UNIT CODE: BIT3208

UNIT NAME: ADVANCED WEB DESIGN AND DEVELOPMENT

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SEMESTER: 3.2

PROJECT TITLE: WEEK 8

SELECTED TECHNOLOGIES: PHP + MySQL

GITHUB LINK: <https://github.com/Coutts007/Advancedweb>

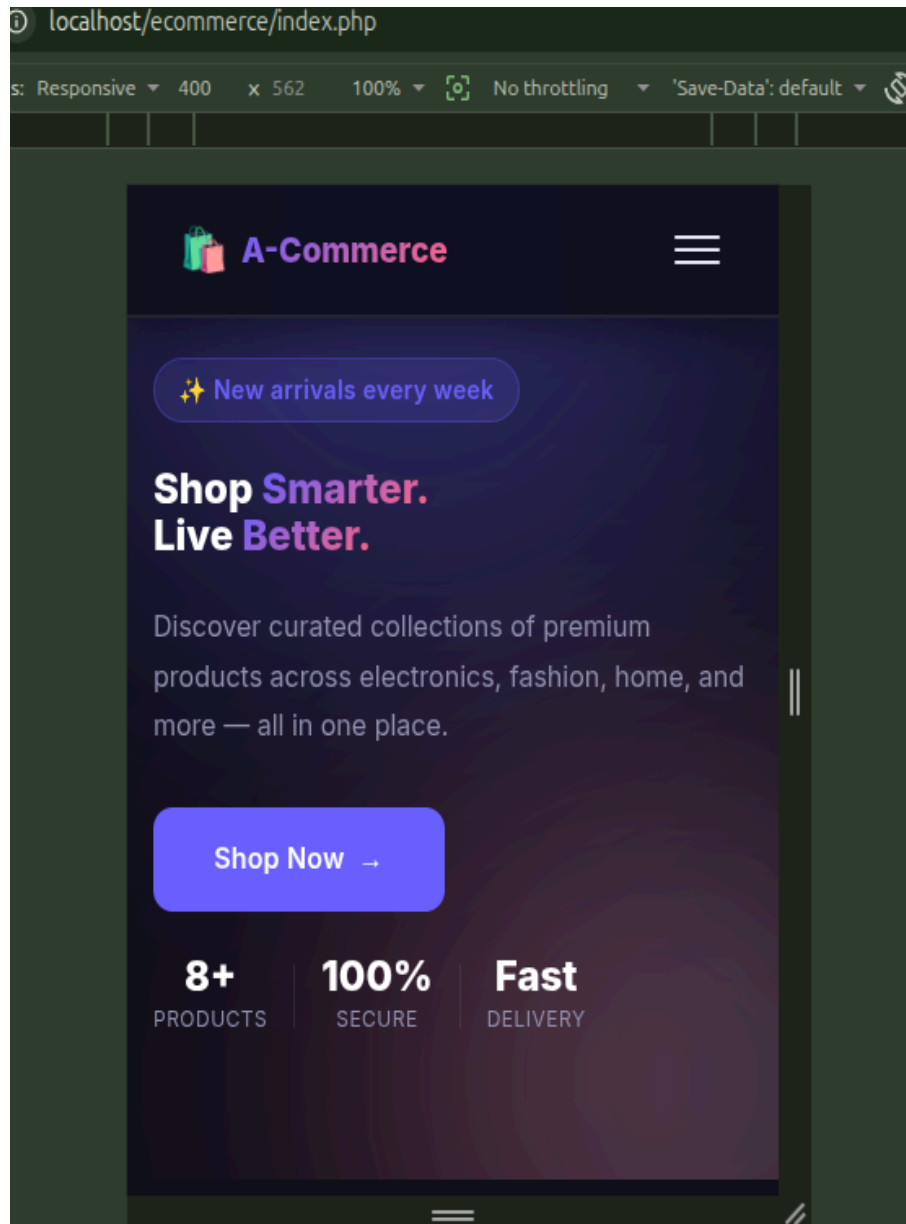
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WEEK 8: RESPONSIVE WEB DESIGN & MOBILE-FIRST DEVELOPMENT

1. Mobile-First Design

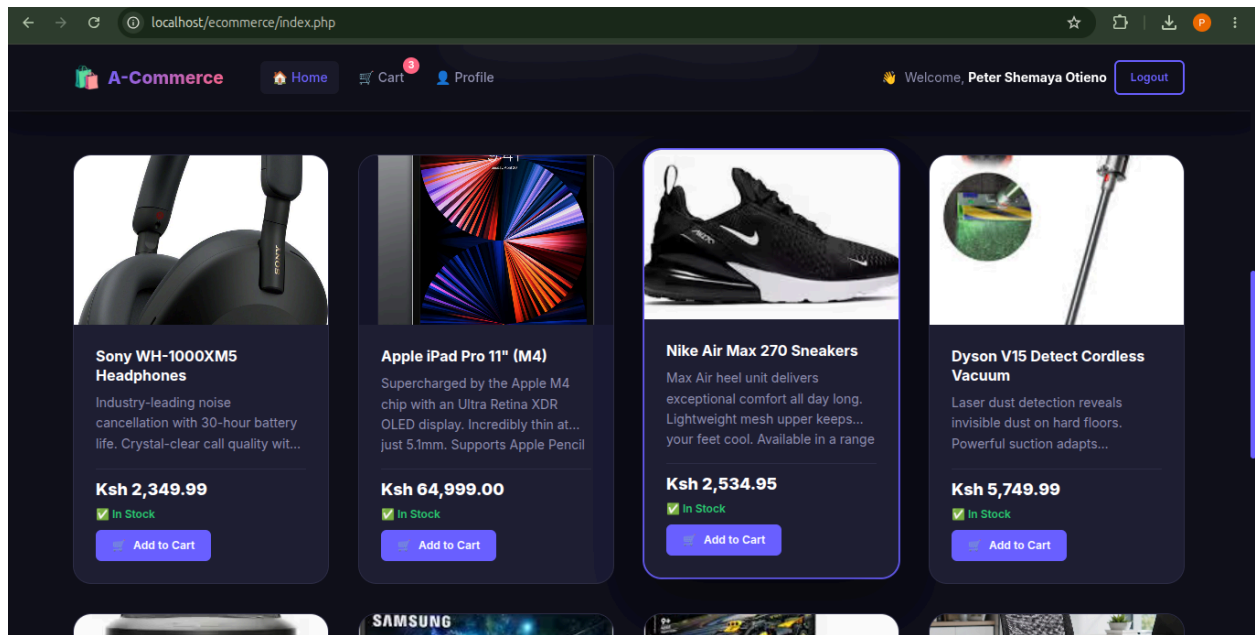
Fig. 1.1: Mobile viewport layout rendering core content layers



Explanation: This interface presents the user view scaled down to a mobile viewport width. Unnecessary clutter is hidden, the main text flows linearly, and primary metrics collapse vertically to match device constraint variables cleanly.

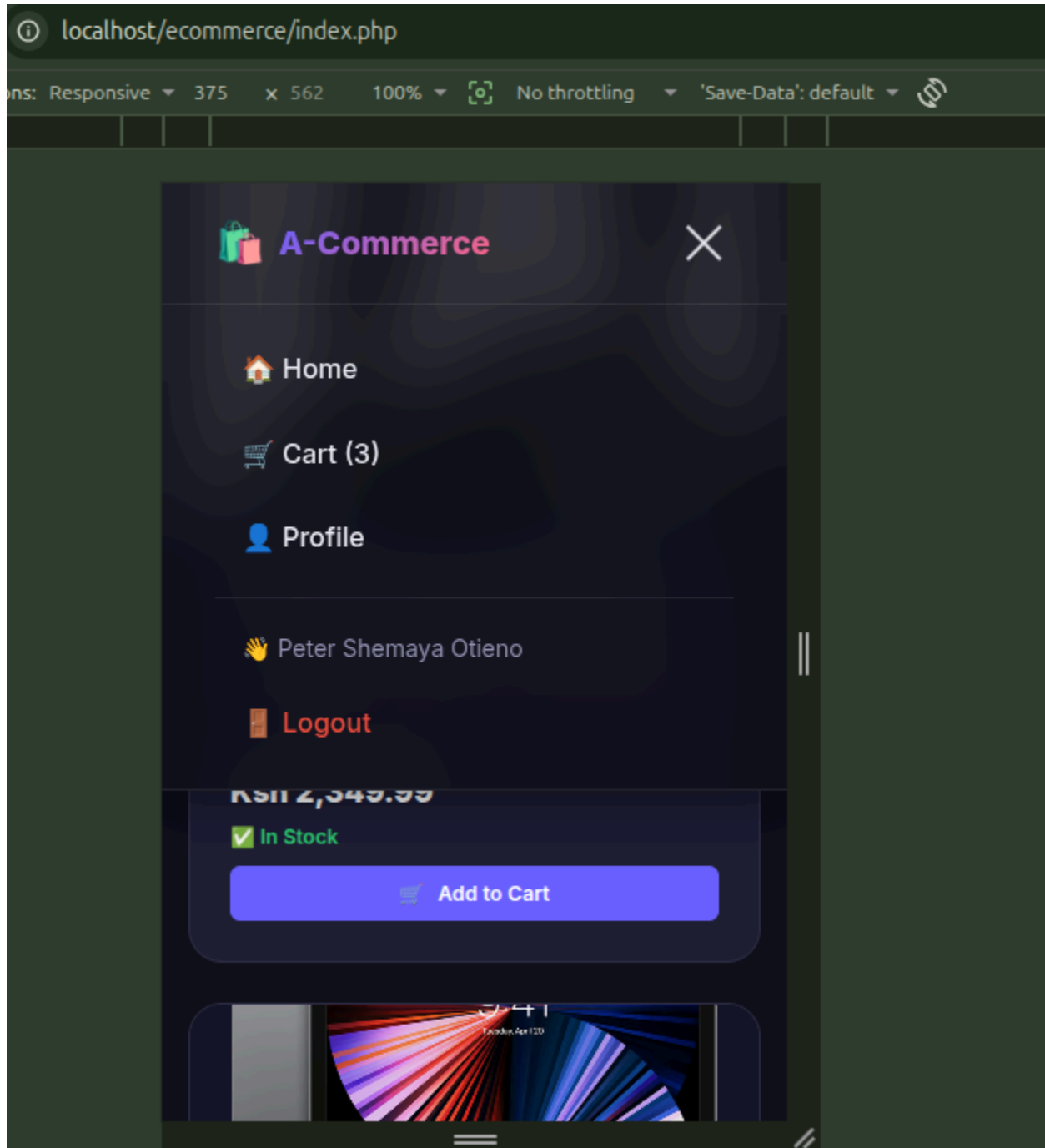
2. Responsive Navigation

Fig. 2.1: Desktop view featuring persistent horizontal navigation link rows



Explanation: This wide-screen render demonstrates the navigation setup under expansive space conditions, utilizing horizontal flex distributions to present links and account profiles across a singular navbar zone.

Fig. 2.2: Mobile menu overlay active following trigger engagement



Explanation: This interface view depicts the active state of an overlay navigation menu layer on a mobile constraint layout, rendering clear access blocks and session logout controllers following interaction.

3. Flexbox and CSS Grid

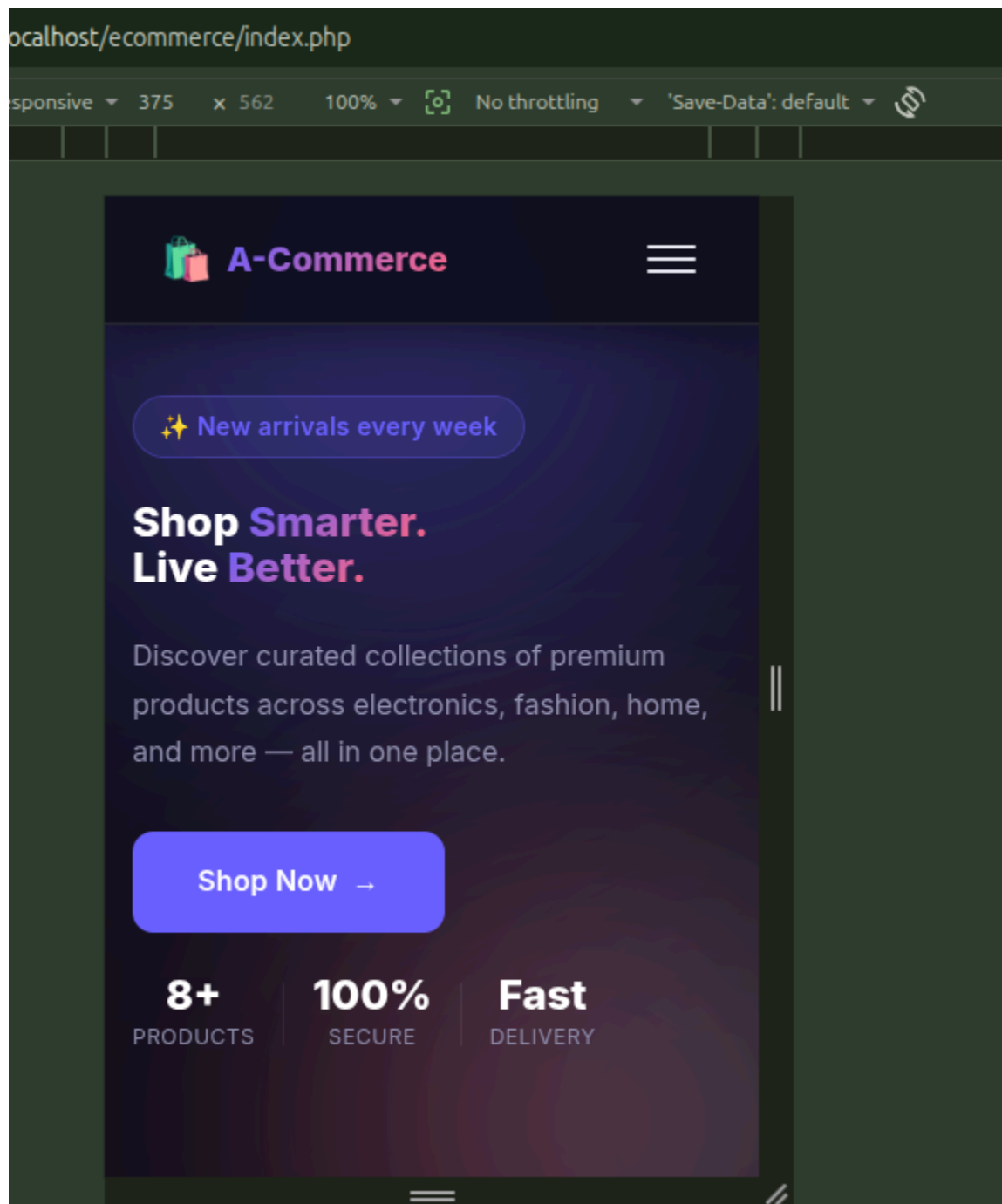
Fig. 3.1: CSS definitions governing layout flow and inner element properties

```
214 .navbar--scrolled {
215     box-shadow: 0 4px 30px rgba(0,0,0,0.5);
216 }
217
218 .navbar-inner {
219     display: flex;
220     align-items: center;
221     gap: var(--space-lg);
222     max-width: var(--container-max);
223     margin-inline: auto;
224     padding: 0 var(--space-xl);
225     height: var(--navbar-h);
226 }
227
228 .navbar-brand {
229     display: flex;
230     align-items: center;
231     gap: var(--space-sm);
232     flex-shrink: 0;
233     transition: opacity var(--transition-fast);
234 }
235 .navbar-brand:hover { opacity: 0.85; }
236
237 .brand-icon { font-size: 1.5rem; }
238 .brand-name {
239     font-size: 1.2rem;
240     font-weight: bold;
241 }
```

Explanation: This code block highlights structural utility rules using flexible display modes, element alignment rules, and custom property configurations to build component layouts.

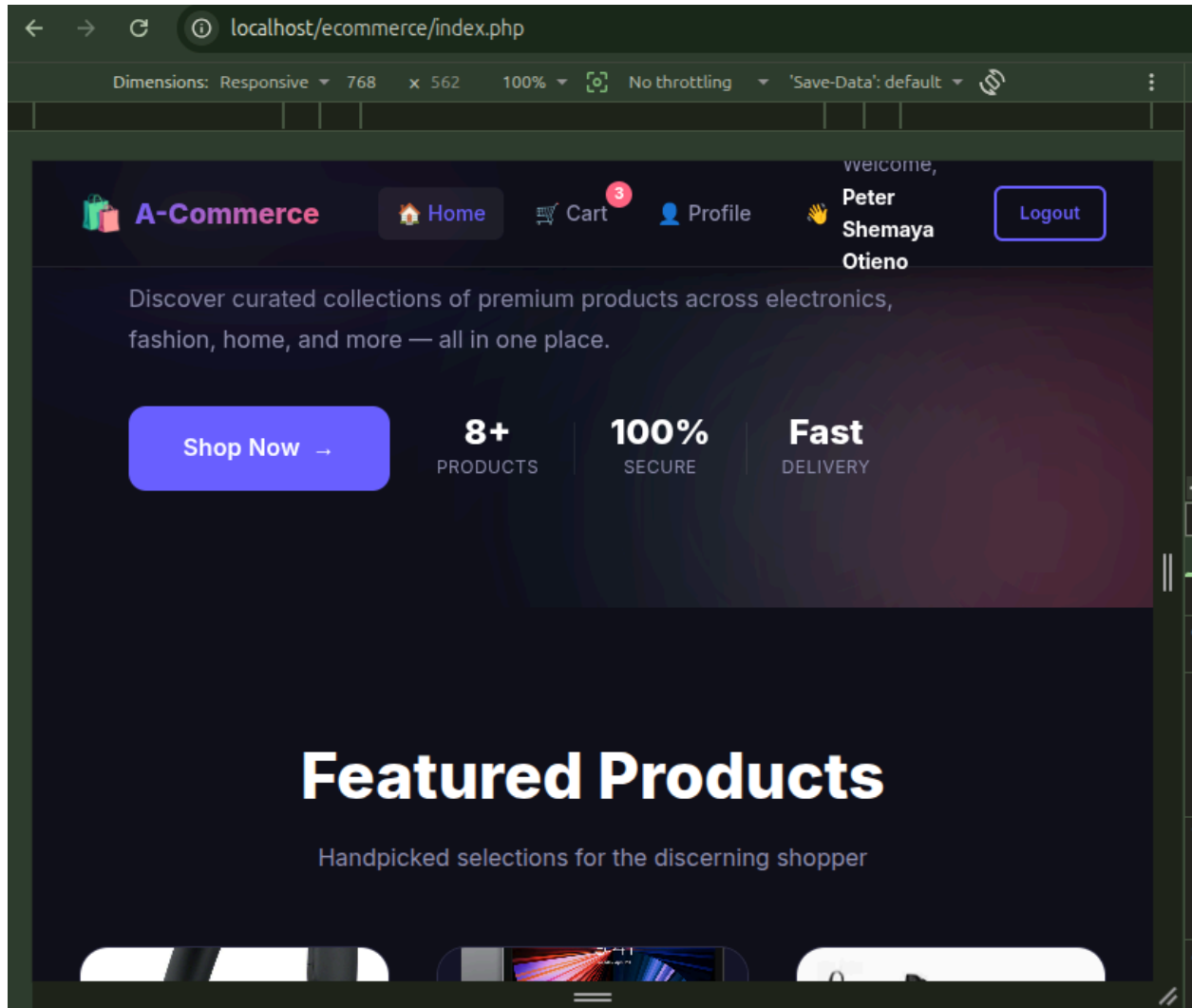
4. Work on Mobile, Tablet, and Desktop Devices

Fig. 4.1: Adaptive layout flow presentation running on mobile view rules



Explanation: This rendering verifies mobile-viewport conformity, isolating high-priority user actions and single-column element grouping rules within thin viewport boundaries safely.

Fig. 4.2: Mid-tier responsive view managing wider design proportions



Explanation: This view highlights mid-tier adaptive adjustment patterns, enabling multi-column flex distributions and proportional title scaling to fill extended space environments smoothly.